[INCH-POUND]
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SUPERSEDING
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24 June 1968

# COMMERCIAL ITEM DESCRIPTION
# SODIUM SULFATE, ANHYDROUS, TECHNICAL (FOR AMMUNITION USE)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This CID covers the general requirements for anhydrous sodium sulfate used for the manufacturing of propellants used in ammunition.

2. SALIENT CHARACTERISTICS.

2.1 Materials. The anhydrous sodium sulfate shall be in the form of a white powder when visually examined and shall be free from lumps, caking, and visual impurities. The anhydrous sodium sulfate shall be provided in the quantity specified in the contract or order (see 6.2).

2.2 Loss on ignition. The moisture content ($H_2O$) of the anhydrous sodium sulfate shall not exceed 0.5 percent when tested in accordance with American Chemical Society (ACS) publication, "Reagent Chemicals". Loss on ignition would include the moisture content ($H_2O$) and other volatiles or trace impurities.

2.3 Assay. The anhydrous sodium sulfate shall not be less than 98.0 percent pure when tested in accordance with American Chemical Society (ACS) Specifications, "Reagent Chemicals".

2.4 Neutrality. The acidity shall be not more than 0.005 percent calculated as sulfuric acid, or the alkalinity shall be not more than 0.5 percent calculated as sodium carbonate when tested in accordance with paragraph 4.2.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army TACOM-ARDEC, Attn.: AMSTA-AR-EDE-S, Bldg 12, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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2.5 Chlorides. The chloride content shall not exceed 0.7 percent by weight when tested in accordance with American Chemical Society (ACS) Specifications, "Reagent Chemicals".

2.6 Insoluble matter. The insoluble matter shall not exceed 0.1 percent by weight when tested in accordance with American Chemical Society (ACS) Specifications, "Reagent Chemicals".

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Neutrality test. Dissolve a 10.00 gm. sample in 200 ml. of carbon dioxide free water and add methyl red indicator. If the solution is acid, titrate to a change in color with 0.01 N sodium hydroxide solution and calculate the acidity as follows (see 2.4):

$$\text{Percent sodium sulfuric acid} = \frac{4.9\ NV}{W}$$

Where: N = normality of sodium hydroxide solution
V = volume of sodium hydroxide solution
W = weight of sample, gm.

If the solution is basic, titrate with 0.01 N hydrochloric acid and calculate the alkalinity as follows:

$$\text{Percent sodium carbonate} = \frac{5.3\ NV}{W}$$

Where: N = normality of hydrochloric acid
V = volume of hydrochloric acid
W = weight of sample, gm.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 The Code of Federal Regulations (CFR) is available by mail order from the Superintendent of Documents, ATTN: New Order, P.O. Box 371954, Pittsburgh, PA 15250-7954.

6.1.2 ACS Specifications publication is available from American Chemical Society, Publications Support Services, 1155 Sixteenth Street, NW, Washington, DC 20036.

6.2 Ordering data. Procurement documents should specify the following:
a. Title, number, and date of this CID.
b. Quantity required (see 2.1).
c. Packaging and marking requirements (see 5).



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6.3 Material Safety Data Sheets. Products meeting the requirements of this CID may be subject to the requirements of 29 CFR 1910.1200 and FED-STD-313 for Material Safety Data Sheets (MSDS).

6.4 Transportation. Anhydrous sodium sulfate is subject to the requirements of 49 CFR 100-185 regarding the transportation of hazardous materials.

6.5 Keywords.
Chemical
Propellant

MILITARY INTEREST:

Custodian
Army-AR

Review Activities:
Army-AV, MD1
DLA-GS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Army-AR

Project 6810-1607



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE *(YYMMDD)* |
|---|---|---|
| | A-A-59279 | 980713 |

**3. DOCUMENT TITLE** Sodium Sulfate, Anhydrous, Technical (For Ammunition Use)

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include ZIP Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) AUTOVON *(If applicable)* | **7. DATE SUBMITTED** *(YYMMDD)* |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE *(Include Area Code)* | |
|---|---|---|
| Mr.Gary Vander Sande | (1) Commercial<br>973 724-2163 | (2) AUTOVON<br>880-2163 |
| c. ADDRESS *(Include ZIP Code)*<br>Commander<br>US Army TACOM-ARDEC<br>ATTN: AMSTA-AR-EDE-S<br>Bldg 12<br>Picatinny Arsenal, NJ 07806-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>**Defense Standardization Program Office, DLSC-LM**<br>8725 John J. Kingman Road, Suite 2533, Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 AUTOVON 427-6888 | |



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